

IDEA PROPOSAL

ITS203 – Object-Oriented Design and Programming Individual Assignment – Milestone 1: Idea Proposal

Student: Safayet Bin Kayes (s2500246)

Project Title: Student Attendance System

1. Problem Description

Many small educational institutions still rely on manual attendance tracking using paper sheets or basic spreadsheets. These methods are slow, prone to errors, difficult to update, and do not provide an organised way to manage multiple classes or students. Teachers need a simple, efficient, and reliable tool to manage class groups and record attendance without depending on large LMS platforms.

This project aims to solve this problem by creating a lightweight Windows Forms application that allows teachers to manage students and record attendance digitally.

2. Proposed Idea

The proposed solution is a **Student Attendance System**, a Windows Forms application built in C#. The system will allow users to:

- Create and manage class groups
- Add and remove students
- Mark attendance as Present, Absent, or Late
- Store attendance records in memory
- Navigate through multiple forms

The application will demonstrate all required OOP principles, including classes, encapsulation, inheritance, polymorphism, abstraction, and exception handling.

3. Inspiration

This idea was inspired by real classroom environments where attendance is still recorded manually. Many small colleges and tutoring centres do not use advanced digital systems,

and teachers often struggle with maintaining accurate records. A simple, user-friendly attendance system can significantly improve efficiency and reduce errors.

I chose this project because it is practical, achievable within the assignment timeframe, and aligns perfectly with the OOP concepts taught in the unit.

4. Functional Requirements

The system should provide support for various essential features that enable a teacher to efficiently handle the classes and keep a track of students' attendance. The application should be able to add classes, add students to a class, delete students from a class (if required), and view all students in a class. It also needs to enable attendance to be recorded with the following three simple choices: Present, Absent, and Late. When attendance is saved, the system should give a message out indicating that it has been saved, otherwise the user has no idea the action has been successful or not. Transitions between forms should be smooth and easy. The application will be developed in terms of user interface, Windows Forms. The layout should be clear and easy to understand, including easy-to-follow labels and controls. Navigation will be accomplished with buttons and selections (student or attendance status) will be made with ListBoxes and ComboBoxes. The data handling will be based on in-memory Lists for the students and attendance records. The interface should update automatically when data is changed, such as when a student is added to the database, or when it is updated to reflect attendance data—so the user always sees current data.

5. Non-Functional Requirements

The system should be user friendly and should have a user friendly interface that can be used by anyone even if they are not technologically savvy. The tasks should have fewer steps possible. Also, performance is critical: forms should load fast; the updates to lists should update immediately without wait. Another important stipulation is reliability. The system should allow only valid input to avoid errors and provide error messages when something goes wrong. To support future maintenance, the codebase should be organised with a clean class structure and include comments where needed. The logic that is used to deal with the data should not be mixed with interface logic. The application should be portable, that is run on any computer running Windows with the .NET installed. It is possible to install and test the basic version of the project without setting up a database.

7. Conclusion

The Student Attendance System is a practical and achievable project that demonstrates all required OOP principles. It solves a real-world problem, includes multiple forms, and provides a clean user experience. The project fits the assessment scope and can be extended with database integration in the future.